



Published in final edited form as:

J Phys Chem B. 2023 April 06; 127(13): 2918–2926. doi:10.1021/acs.jpcb.2c08812.

Spatiotemporal heterogeneity of de novo lipogenesis in fixed and living single-cells

Sydney O. Shuster,

Michael J. Burke,

Caitlin M. Davis*

Department of Chemistry, Yale University, 225 Prospect Street, New Haven, CT 06511, USA

Abstract

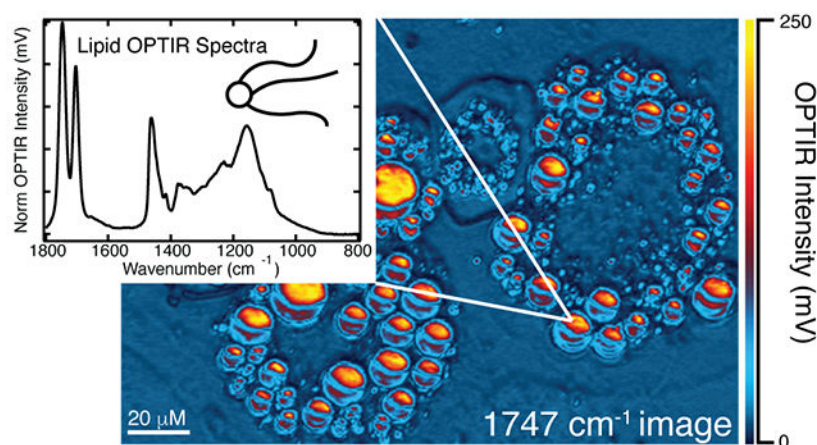
De novo lipogenesis (DNL) is a critical metabolic process that provides the majority of lipids for adipocyte and liver tissue. In cancer, obesity, type II diabetes, and nonalcoholic fatty liver disease DNL becomes dysregulated. A deeper understanding of the rates and of sub-cellular organization of DNL is necessary for identifying how this dysregulation occurs and varies across individuals and diseases. However, DNL is difficult to study inside the cell because labeling lipids and their precursors is not trivial. Existing techniques either can only measure parts of DNL, like glucose uptake, or do not provide spatiotemporal resolution. Here, we track DNL in space and time as isotopically labeled glucose is converted to lipids in adipocytes using optical photothermal infrared microscopy (OPTIR). OPTIR provides sub-micron resolution infrared imaging of the glucose metabolism in both living and fixed cells while also reporting on the identity of lipids and other biomolecules. We show significant incorporation of the labeled carbons into triglycerides in lipid droplets over the course of 72 hours. Live cells had better preservation of lipid droplet morphology but both showed similar DNL rates. Rates of DNL, as measured by the ratio of ^{13}C labeled lipid to ^{12}C labeled lipid, were heterogeneous, with differences within and between lipid droplets and from cell to cell. The high rates of DNL measured in adipocyte cells match upregulated rates of DNL previously reported in PANC1 pancreatic cancer cells. Taken together, our findings support a model where DNL is locally regulated to meet energy needs within cells.

Graphical Abstract

*Corresponding Author: c.davis@yale.edu. Caitlin M. Davis. Department of Chemistry, Yale University, 225 Prospect Street, New Haven, CT 06511, USA.

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.



INTRODUCTION

De novo lipogenesis (DNL) is a critical metabolic process, producing the majority of lipids in adipose and hepatic tissue.¹ Dysregulation of the pathway is closely linked to obesity, nonalcoholic fatty liver disease, and type II diabetes and the pathway is known to be upregulated in cancer, likely as a way to maintain glycolytic flux.^{2,3} DNL is difficult to study as lipids can be challenging to label, especially with fluorescent labels like dyes and fluorescent proteins that have been used extensively to study other metabolic processes. Additionally, glucose based metabolic probes are often not utilized in the same way as natural glucose.^{4–6} Labeling glucose with ¹³C, however, has been shown to be nontoxic and nonperturbative to cells and is anabolized like ¹²C glucose.⁷ Isotopic labeling has been used to great success for diverse metabolic studies, including mass-spectrometry, but these methods lack spatial resolution within the cell.^{7,8} Recent research has highlighted how heterogeneous metabolism can be across the cell and so spatial resolution is critical for a full understanding of how metabolism changes in response to stress and disease.^{9–11}

Vibrational microspectroscopy is a promising tool for studying metabolism in cells and organisms as it is label-free and non-perturbative.^{12–15} It can provide spatially and temporally resolved information on the identity of biomolecules in the cell as well as information on the secondary structure of protein molecules.¹⁶ Additionally, when small vibrationally active probes such as azide, deuterium, or ¹³C labels are used, vibrational microspectroscopy can inform on site-specific local environment and rates of metabolism across the cell.¹⁷ For example, Li and Cheng used deuterium labeled glucose for tracking DNL in live cancer cells using Raman microspectroscopy.⁴ However, their method could not also provide information on lipid identity and other surrounding biomolecules as it was based on measurements at a single wavenumber.

The two primary forms of vibrational microspectroscopy used for cellular studies are FTIR and Raman spectroscopy. Unfortunately, both methods have significant drawbacks affecting in cell work. FTIR suffers from poor spatial resolution when used with microscopy.¹⁸ Depending on the wavelength, FTIR has a diffraction limited spatial resolution ranging from ~3 μm to greater than 20 μm. This obscures all but the largest of cellular organelles and

substructures. Additionally, FTIR cannot easily be performed in aqueous media, as water is a strong infrared absorber.^{19,20} This means that work is usually done on cells that have been fixed and dried,^{13,17,21} possibly disrupting the cellular environment as water is removed,²² introducing artifacts such as Mie scattering,²³ and not allowing for real time, live cell tracking of cellular processes. Raman spectroscopy, on the other hand, has significantly less interference from water and can be performed on live cells.^{4,21,22,24–26} However, the Raman cross-section is small and signal intensity and signal to noise are consistent issues.²⁷ This necessitates high laser powers and long collection times, which can damage cells.

Optical photothermal infrared spectroscopy (OPTIR) was developed to address some of these issues.^{25,28,29} In OPTIR, the sample is ‘pumped’ with an IR laser, and rather than directly measuring IR absorbance, the sample is ‘probed’ with a visible laser. The IR absorbance causes photothermal heating and expansion, which in turn changes the refractive index of the measured spot. This affects the intensity of the probe beam, which can be collected in either reflection or transmission mode. This data is amplified and used to reconstruct the IR spectrum of the sample. Because the measurement is made by the visible beam, the diffraction limit is based on the visible laser, bringing the spatial resolution in line with traditional optical microscopy at sub-500 nm. Since OPTIR is not an absorbance based technique, water also contributes less strongly to the total signal, allowing for measurements in cells.²⁵ OPTIR has been performed on fixed and live cell data at high resolution^{25,29,30}, but its ability to track metabolic processes and make use of vibrational probes has not yet been demonstrated.

In this work, we use OPTIR to investigate the rate of DNL in differentiated 3T3-L1 adipocytes. In DNL, glucose or other carbohydrates are metabolized to form free fatty acids (FFA) and triglycerides (Fig. 1).¹ The pathway begins with glycolysis. Pyruvate is then taken into the mitochondria and converted to citrate via the TCA cycle. Citrate is converted to acyl-CoA and malonyl-CoA which feed the fatty acid synthesis pathway and result in triglycerides which are then stored in lipid droplets as an energy reserve. By feeding cells ¹³C labeled glucose, the ¹³C progresses through the DNL pathway and is incorporated into the resulting lipids, enabling tracking of DNL rates. We examine both fixed and living cells and use ratio images to determine varying rates of DNL across different regions of the cell. We observe heterogeneity in the rates of DNL across the cell and especially between cells. This work proves the spatial and spectral resolution of OPTIR and its sensitivity to live cell work in aqueous media, opening the door to tracking of other metabolic processes in healthy and disease states.

MATERIALS AND METHODS

Materials.

Unless otherwise specified, reagents were sourced from Sigma-Aldrich.

Cell culture, ¹³C labeling, and cell fixing.

3T3-L1 cells (ATCC, Manassas, VA) were cultured in 10-cm diameter sterile petri dishes (Corning, Corning, NY). Low passage number (<10), pre-differentiated cells were grown

to confluence in Dulbecco's modified Eagle's medium containing 4.5 g/liter glucose and L-glutamine (DMEM, Corning) supplemented with 10% calf bovine serum (CBS, ATCC) and 1% penicillin/streptomycin (Thermo-Fischer, Waltham, MA) under standard conditions (37 °C, humidified atmosphere, 5% CO₂). Two days post confluence, medium was changed to DMEM containing 10% fetal bovine serum (FBS) (Corning), 1% penicillin/streptomycin, 20 µg/ml insulin, 250 nM dexamethasone and 500 µM isobutylmethylxanthine. After 2-3 days, cells were trypsinized and replated on CaF₂ coverslips (20 x 20 x 0.35 mm) (Crystran, Poole, UK) in 50 mm dishes (Corning) and medium was changed to DMEM with FBS, antibiotic, and 20 µg/ml insulin (post-differentiation medium). Media was exchanged every 2-3 days until lipid droplets formed and stabilized (5-7 days). At this point medium was changed to post-differentiation media without glucose supplemented with ¹³C labeled glucose (Cambridge Isotope Laboratories, Tewksbury, MA) to 4.5 mg/mL. At each desired time point cells were fixed with 4% paraformaldehyde (PFA) in PBS (Corning) for 20 minutes at room temperature, washed three times with PBS, three times with MilliQ purified water, and allowed to air dry before analysis.

Live cell chamber construction.

Cells imaged live were grown and differentiated as stated above. 72 hours after the addition of ¹³C glucose, cells were washed three times with PBS and mounted in PBS on a glass microscope slide (VWR, Radnor, PA). Double sided tape spacers (Nitto, San Diego, CA) with a 5 µm thickness were used to keep cells hydrated in PBS, create an airtight seal, and minimize compression of the cell.

OPTIR data collection.

Imaging was performed on a mIRage IR microscope (Photothermal Spectroscopy Corporation, Santa Barbara, CA) integrated with a four-module-pulsed quantum cascade laser (QCL) system (Daylight Solutions, San Diego, CA) with a tunable range from 1799 to 801 cm⁻¹. Brightfield optical images were acquired using a low magnification 10× refractive objective, with a working distance of 15 mm. OPTIR spectra and images were collected with a high magnification (IR) 40×, 0.78 NA (8 mm working distance) all reflective Cassegrain objective (PIKE Technologies, Fitchburg, WI). Fixed cell spectra were collected in standard (reflective) mode and live cell were collected in transmission mode. Spectra were collected and processed using PTIR Studio 4.4 (Photothermal Spectroscopy Corporation). The laser powers of the QCL IR Pump and green (532 nm) probe laser were set to 10 and 4.8%, respectively. These laser powers were selected because they provided the best signal to noise and did not damage fixed cells during long collection times. The laser pulse rate was 100 kHz with a duty cycle of 5%, a pulse width of 500 nm, and a gain of 5× for fixed samples and 1× for live samples. The hyperspectral map size for the fixed cells has step size of 500 nm, the maximum resolution of the instrument. All OPTIR spectra were collected one scan unless otherwise indicated.

Data Analysis.

Fixed and live cell data was analyzed using PTIR Studio 4.4 and images processed using Fiji (NIH, Bethesda, MD).³¹ Live cell ratio images were generated using Python 3.7 in Colab (Google, Mountain View, CA).³² Fixed cell single spectra were analyzed using IGOR Pro

8 (Wavemetrics, Portland, OR). Fixed cell hyperspectral maps were vector normalized and analyzed using Python 3.7 in Colab. Fixed cell $^{13}\text{C}/^{12}\text{C}$ ratio images were corrected for the amide-I intensity using the eq 1.:

$$\frac{^{13}\text{C}}{^{12}\text{C}}_{\text{ester carbonyl intensity}} = \frac{\bar{A}_{1693-1713} - 0.31 \times \bar{A}_{1645-1655}}{\bar{A}_{1737-1757}}$$

where $\bar{A}$ is the average signal intensity over the indicated frequencies. $\bar{A}$ was calculated from full spectra of fixed, control adipocytes. Fixed cell ratios (eq. 1) were only performed on regions of the cell that had lipid, defined as a ^{12}C lipid ester carbonyl intensity of 0.025 or greater after vector normalization. Live cell ratio images were constructed from single wavenumber images and were corrected for both water and the amide-I band intensity using the following eq. 2:

$$\frac{^{13}\text{C}}{^{12}\text{C}}_{\text{ester carbonyl intensity}} = \frac{\bar{A}_{1693-1713} - 0.66 \times \bar{A}_{1645-1655}}{\bar{A}_{1737-1757}}$$

Where $\bar{A}$ is the average signal intensity over the indicated frequencies. This was calculated from full spectra of live, control adipocytes.

RESULTS AND DISCUSSION

^{13}C is a nonperturbative vibrational probe of *de novo* lipogenesis.

To assess the power of OPTIR in resolving cellular components, the spatial and spectral resolution was evaluated using fixed 3T3-L1 cells. The cells were grown to confluency and differentiated using standard practices.³³ The resulting lipid droplet rich, adipocyte cells were fixed using PFA and air dried to limit background from water.

The spectral resolution was evaluated in a representative spectra collected in a lipid deposit of a fixed adipocyte cell (Fig. 2a). All peaks in the adipocyte OPTIR spectra could be assigned to cellular components. All bands, except the peaks at 1655 cm^{-1} and 1541 cm^{-1} , the amide-I and amide-II bands, respectively¹⁶, were assigned to lipid vibrational modes.^{24,34-36} Notably, the position of the ester carbonyl vibrational mode at 1747 cm^{-1} indicates that the bulk of the lipids present are triglycerides, as expected for lipid droplet rich adipocytes.^{33,35} In spectra collected in regions with low lipid content, such as the nucleus of the cells, the phosphate stretches of nucleic acids are visible (Fig. S1).

To track glucose metabolism and DNL, ^{13}C glucose was fed to cells in place of ^{12}C glucose. Because the frequency of the vibrational modes of molecules has an inverse relationship with the reduced mass of the atoms involved, substituting heavier isotopes cause red-shifts in the frequency of the modes.^{37,38} Glucose is the primary carbon source for generation of FFA and triglycerides in adipocytes like 3T3-L1 (Fig. 1).³⁹ Incorporation of the ^{13}C into lipid molecules results in significant shifts across the lipid spectra (Fig. 2b).^{17,40} The most obvious is the $\sim 44\text{ cm}^{-1}$ red shift of the ^{13}C ester carbonyl stretch as compared to the ^{12}C ester carbonyl stretch. As the single strongest lipid vibrational mode, the ester carbonyl

stretch of triglycerides is an excellent probe of lipids within the cell. By calculating the ratio of the ester $^{13}\text{C}=\text{O}$ band to the ester $^{12}\text{C}=\text{O}$ over time, the rate of DNL can be tracked.¹⁷

Significant shifts are also observed in the CH_3 deformation and the symmetric and asymmetric C-O-C and C-O stretches. Interestingly, the band at 1239 cm^{-1} , which could not be unambiguously assigned to C-O-C or PO_2^- asymmetric stretch in the unlabeled cell, disappears upon ^{13}C labeling, likely red shifting into the neighboring peaks. This suggests that the primary source of that peak is the C-O-C asymmetric stretch of triglycerides and/or cholesteryl esters and that phospholipids do not significantly contribute to the signal in 3T3-L1 adipocytes.

Some bands do not shift after ^{13}C glucose feeding. As anticipated, bands associated with proteins are unchanged. For example, the two strong bands centered at $\sim 1650\text{ cm}^{-1}$ and $\sim 1540\text{ cm}^{-1}$ that are attributed to the amide-I and amide-II modes of proteins do not shift upon ^{13}C glucose feeding. Surprisingly, the band at $\sim 1450\text{ cm}^{-1}$ associated with CH_3 and CH_2 deformations and stretching of proteins and lipids are unchanged. This may be because triglycerides are formed from the esterification of FFA and glycerol. Although both FFA and glycerol are produced from glucose via glycolysis, they are produced and recycled at different rates, affecting the rates of shifts of those bands.

OPTIR provides high resolution hyperspectral imaging.

The spatial resolution of OPTIR is determined by the diffraction limit of the 532 nm probe laser. Line scans demonstrate high spatial resolution (Fig. S2); OPTIR has a spatial resolution of 500 nm.²⁹ Hyperspectral maps are three-dimensional images assembled from complete OPTIR spectra, $801\text{--}1799\text{ cm}^{-1}$, collected at every point. Figure 3 shows a brightfield image (Fig. 3a) and vibrational images of a control adipocyte generated from the ^{12}C ester carbonyl lipid band (Fig. 3b) and the amide-I protein band (Fig. 3c). The lipid signal is clustered in lipid droplets and deposits and there is limited overlap between the protein and lipid signal.

Noticeably, as compared to the appearance of living adipocytes, fixed, dried cells experienced some pooling and combining of the lipid droplets into deposits. This has been previously reported and does not affect the lipid spectra, but does affect lipid distribution and overlap with protein signal.³⁵

Spatiotemporal resolution of DNL in fixed cells.

To track DNL, hyperspectral maps of dried, fixed cells at 24, 48, and 72 hour time points after the addition of ^{13}C glucose were collected (Fig. 4). All cells had a similar appearance with collections of lipid droplets and deposits that could be visualized by the intensity of the ^{12}C carbonyl ester band, and protein signal throughout the rest of the cell, visualized via the intensity of the amide-I band.

Ratio images of the ester $^{13}\text{C}=\text{O}$ band to the ester $^{12}\text{C}=\text{O}$ map DNL were constructed to quantify DNL in each cell (Fig. 4 ^{13}C incorporation). The images were assessed in two ways, as a single-cell average of the lipid rich areas or with full spatial resolution over the

lipid rich areas. We chose only to analyze the regions with significant lipid signal to reduce noise in lipid free areas that may be amplified by creating ratio images.

We found that the average single-cell $^{13}\text{C}/^{12}\text{C}$ ratio continually increased over the period of time tested (72 hours) with a final ratio of 0.54 ± 0.14 (Fig. 5a and Table S1). However, there were significant variations in the incorporation of ^{13}C into the triglycerides. The large standard deviation in the average ratios (Fig. 5a) highlights cell to cell difference in average rates of DNL. DNL is also spatially heterogeneous; at 72 h, histograms of the $^{13}\text{C}/^{12}\text{C}$ ratios measured in lipid rich regions of single cells vary from 0 to above 1, indicating that the local ^{13}C enrichment ranges from 0 to well above 50% (Fig. 5b–d, Table S1). Further incorporation of ^{13}C would likely occur after 72 h, but we chose this as the end point as it was in line with previous literature^{4,17} and longer incubations would have required a secondary feeding of the cells which may cause disruption. The data broadly agrees with two past studies that tracked DNL via heavy isotope incorporation into triglycerides. Shi *et al.* used QCL-based FTIR imaging to obtain ratios of ^{13}C lipid ester carbonyl/ ^{12}C lipid ester carbonyl in 3T3-L1 MBX cells at 72 h with low spatial resolution. They did not report any average data, however, the ^{13}C lipid ester carbonyl/ ^{12}C lipid ester carbonyl ratio range reported in their cell images (ratios between 0.1 – 0.7) agrees with the ranges we see. Our measurements are also in line with DNL rates reported for PANC1 pancreatic cancer cells; Li and Cheng reported end points of DNL at 72 h (a ratio of ~ 0.6) via a ratio of the lipid C-D/ lipid C-H as measured by single frequency SRS imaging.^{4,17} Normal immortalized epithelial cells and prostate cancer cell lines, LNCaP and PC3, had significantly lower ratio end points.⁴ To meet increased energy needs, cancer cells rely on metabolic rewiring such as upregulation of DNL.¹ It is intriguing that a similar rate of DNL is observed in 3T3-L1 adipocyte and PANC1 pancreatic cancer cells, as it suggests that DNL is so upregulated in this particular cancer cell line that it is in line with adipocytes.

In line with the single-cell average data (Fig. 5), as time progressed the ratio of $^{13}\text{C}=\text{O}/^{12}\text{C}=\text{O}$ labeled lipids within the cell increased in the ratio image (Fig. 4 ^{13}C incorporation). The spatial heterogeneity of ^{13}C is evident in these images. In particular, the edges of lipid droplets and deposits have a lower ratio of ^{13}C labeling, especially at later time points (Fig. 4c), suggesting the presence of older lipids or slower DNL in these locations. It is important to note that these lower $^{13}\text{C}/^{12}\text{C}$ ratios are not due to decreased concentrations or pathlengths at the edge, because both artifacts are corrected for in the ratio. Nevertheless, it is difficult to draw strong conclusions about the spatial heterogeneity of DNL from ratio images as the shape and structure of lipid droplets is not well-preserved during fixing and drying

To better understand subtle intracellular spectral differences, spectral clustering was performed on the ester carbonyl and amide-I region of each hyperspectral map (Fig. S3). Cluster analysis sorts the spectra into groups of high of similarity based only on spectral features^{41,42} and separated some clusters with a small shoulder at 1727 cm^{-1} , which is likely indicative of FFA (Fig S3b).³⁵ This shoulder appeared in control cells as well, demonstrating that it is not an artifact of the ^{13}C glucose feeding (Fig. S4). FFA are both precursors to and metabolites of triglycerides and are expected to exist in high quantities in adipocytes.⁴³

Live cell imaging highlights spatiotemporal DNL heterogeneity.

Although fixed, dried cells are advantageous because they enable the long imaging times necessary for collection of highly detailed hyperspectral maps, they also come with several drawbacks. Dried cells can be damaged by IR and visible lasers and drying may introduce artifacts.²³ More importantly, the fixing and drying process causes disruptions in the lipid droplets and possibly in protein structure and localization.^{21,35} Loss of water can disturb organelles and change the shape and thickness of the cell, compromising the quality of spatial data.²¹ Post fixatives can help preserve morphology, but may introduce additional vibrationally active compounds. For example, the most common post-fixative for lipid preservation is OsO₄, which reacts with unsaturated lipids, removing double bonds and crosslinking lipids.^{22,44} OsO₄ fixation has also been known to displace proteins and carbohydrates and cause DNA clumping. These changes would lead to shifts in the vibrational profile. Disruption of lipid droplet morphology is a significant drawback of fixed, dried cells and highlights the importance of live cell data collection to confirm observations made in fixed cells as well as track processes of interest in real time. Therefore, comparing against live cell data is critical to confirm observations made in fixed cells as well as tracking processes of interest in real time. Unfortunately, collecting data from hydrated samples has been nearly impossible using conventional FTIR microscopes as water is a strong IR absorber.²⁰ OPTIR, however, has significantly less contribution from bulk water and data can be collected in live cells.²⁵

Adipocytes at 72 hours after ¹³C glucose feeding were mounted on slides in PBS with a 5 μm spacer to preserve hydration and prevent cell compression. In the current configuration, hyperspectral maps could not be collected because of the long imaging times. Fixed cell images were used to identify frequencies of interest. Single wavenumber images of single cells were collected at 1747 cm⁻¹ (ester ¹²C=O), 1703 cm⁻¹ (ester ¹³C=O), and 1655 cm⁻¹ (water and amide-I bands) (Fig. 6, Fig. S5).

Brightfield images of the live adipocyte cells show distinct lipid droplets and morphology in agreement with literature (Fig. 6a).³³ The 1747 cm⁻¹ images clearly show strong lipid signals in the droplets with no lipid signal in the rest of the cell (Fig. 6b). Due to their low local concentrations, single lipid bilayers are not detectable by IR. Similar to FFA, even if single lipid bilayers were detectable inside cells, the carbonyl of the phospholipids would absorb at a slightly different wavelength than the carbonyl of the triglycerides in lipid droplets. Ratio images show varied rates of DNL, especially between cells (Fig. 6c, Fig. S5). There is, however, some variation between lipid droplets within cells and even across lipid droplets. This hints at the complexity of glucose anabolism into lipids and lipolysis and demonstrate that this technique can detect even small differences in the ratio of ¹³C triglycerides to ¹²C triglycerides in living cells. Other than the shape of the lipid droplets, the data is in good agreement with the fixed cell ratios at 72 h with the ¹³C/¹²C lipid ester carbonyl ratio in lipid rich regions varying from ~0.4 to ~1 in both cases.

Single spectra were collected to compare with fixed cells (Fig. S6). Full spectra show good agreement with fixed cell data and similar ratios between the ¹³C and ¹²C ester carbonyls at the 72 h time point (Fig. 7). The live cell lipid droplets were better defined in spectral images of live cells than in fixed cells and had less overlap with protein amide-I signal, as

seen by full spectra (Fig. S6). While the vibrational mode of water at $\sim 1645\text{ cm}^{-1}$ could obscure the amide-I mode, the absence of proteins in the lipid droplets is confirmed by the absence of the amide-II band (Fig. 7, Fig. S6).^{16,20} This data implies that fixed cells provide reliable data on the range of rates of DNL seen across the cell. Unfortunately, the collapse of the lipid droplets in fixed cells causes protein and lipid signals to overlap, which does not occur to a significant extent in the live cells. This may obscure signal from the lipids and also means that it is impossible to comment on lipid localization in the fixed cells. Therefore, for questions in which high resolution spatial information is needed, live cells should be used. Fixed cells, however, are suitable for hyperspectral imaging, which can provide information on lipid species occurring at lower concentrations, like FFA, and allow complete analysis of lipid and protein spectral features at each location. In the future, controlled temperature stages and flow cells will allow for the collection of OPTIR data in live cells for longer time periods and with changing conditions, allowing for real-time tracking of DNL. Moving forward, this technique can be used to map rates of DNL across many cell lines and in stress and disease states providing further insight on how DNL contributes to metabolic disorders. Further, it can be used to assess how effective drugs that target parts of the DNL pathway are at regulating DNL and treating disease.

CONCLUSIONS

In this work, we track DNL in fixed and live adipocytes via metabolic labeling. Rates of DNL were in line with previous literature and the spatial resolution allowed us to comment on differing rates of DNL across and between cells. The flexibility of OPTIR means that it can be used for both hyperspectral fixed cell imaging and single frequency live cell imaging. The two modalities provided similar results when calculating DNL, but there were significant benefits and drawbacks for both fixed and live single cell imaging. Fixed cell hyperspectral imaging provides the most complete assessment of the biomolecules present in the cell and can even provide information on protein secondary structure across the cell. However, it does not provide an accurate snapshot of the location of lipids within the cell. Live cell single-frequency OPTIR imaging showcased high spatial resolution and good preservation of cell morphology and biomolecule localization, but not the information found in full mid IR hyperspectral imaging. The techniques can be used in tandem to extract as much information as possible on metabolic rates across the cell and can better inform on DNL in healthy and disease states.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

ACKNOWLEDGMENT

SOS was partially supported by the National Science Foundation Graduate Research Fellowship under Grant No. DGE-2139841 and the National Institutes of Health under Biophysics Training Grant No. T32 GM 008283. Financial support for this publication results from a Scialog program sponsored jointly by Research Corporation for Science Advancement and the Gordon and Betty Moore Foundation.

REFERENCES

- (1). Ameer F; Scandiuizzi L; Hasnain S; Kalbacher H; Zaidi N De Novo Lipogenesis in Health and Disease. *Metab. - Clin. Exp* 2014, 63 (7), 895–902. [PubMed: 24814684]
- (2). Mullen AR; Wheaton WW; Jin ES; Chen P-H; Sullivan LB; Cheng T; Yang Y; Linehan WM; Chandel NS; DeBerardinis RJ Reductive Carboxylation Supports Growth in Tumour Cells with Defective Mitochondria. *Nature* 2011, 481 (7381), 385–388. [PubMed: 22101431]
- (3). Gatenby RA; Gillies RJ Why Do Cancers Have High Aerobic Glycolysis? *Nat. Rev. Cancer* 2004, 4 (11), 891–899. [PubMed: 15516961]
- (4). Li J; Cheng J-X Direct Visualization of De Novo Lipogenesis in Single Living Cells. *Sci. Rep* 2014, 4 (1), 6807. [PubMed: 25351207]
- (5). Zou C; Wang Y; Shen Z 2-NBDG as a Fluorescent Indicator for Direct Glucose Uptake Measurement. *J. Biochem. Biophys. Methods* 2005, 64 (3), 207–215. [PubMed: 16182371]
- (6). Yamada K; Nakata M; Horimoto N; Saito M; Matsuoka H; Inagaki N Measurement of Glucose Uptake and Intracellular Calcium Concentration in Single, Living Pancreatic Beta-Cells. *J. Biol. Chem* 2000, 275 (29), 22278–22283. [PubMed: 10748091]
- (7). Metallo CM; Walther JL; Stephanopoulos G Evaluation of ¹³C Isotopic Tracers for Metabolic Flux Analysis in Mammalian Cells. *J. Biotechnol* 2009, 144 (3), 167–174. [PubMed: 19622376]
- (8). Metallo CM; Gameiro PA; Bell EL; Mattaini KR; Yang J; Hiller K; Jewell CM; Johnson ZR; Irvine DJ; Guarente L; et al. Reductive Glutamine Metabolism by IDH1 Mediates Lipogenesis under Hypoxia. *Nature* 2012, 481 (7381), 380–384.
- (9). Park CK; Horton NC Structures, Functions, and Mechanisms of Filament Forming Enzymes: A Renaissance of Enzyme Filamentation. *Biophys. Rev* 2019, 11 (6), 927–994. [PubMed: 31734826]
- (10). Boeynaems S; Alberti S; Fawzi NL; Mittag T; Polymenidou M; Rousseau F; Schymkowitz J; Shorter J; Wolozin B; Bosch LVD; et al. Protein Phase Separation: A New Phase in Cell Biology. *Trends Cell Biol.* 2018, 28 (6), 420–435. [PubMed: 29602697]
- (11). Prouteau M; Loewith R Regulation of Cellular Metabolism through Phase Separation of Enzymes. *Biomolecules* 2018, 8 (4), 160. [PubMed: 30513998]
- (12). Matthäus C; Bird B; Miljković M; Chernenko T; Romeo M; Diem M Infrared and Raman Microscopy in Cell Biology. *Methods Cell Biol.* 2008, 89, 275–308. [PubMed: 19118679]
- (13). Naumann D Infrared Spectroscopy of Cells, Tissues, and Biofluids. In *Encyclopedia of Biophysics*; Roberts GCK, Ed.; Springer: Berlin, Heidelberg, 2013; pp 1057–1065.
- (14). Romeo MJ; Boydston-White S; Matthäus C; Miljković M; Bird B; Chernenko T; Diem M Vibrational Microspectroscopy of Cells and Tissues. In *Biomedical Vibrational Spectroscopy*; John Wiley & Sons, Ltd, 2008; pp 121–152.
- (15). Diem M; Romeo M; Boydston-White S; Miljkovic M; Matthaues C A Decade of Vibrational Micro-Spectroscopy of Human Cells and Tissue (1994-2004). *The Analyst* 2004, 129 (10), 880–885. [PubMed: 15457314]
- (16). Barth A Infrared Spectroscopy of Proteins. *Biochim. Biophys. Acta BBA - Bioenerg* 2007, 1767 (9), 1073–1101.
- (17). Shi L; Liu X; Shi L; Stinson HT; Rowlette J; Kahl LJ; Evans CR; Zheng C; Dietrich LEP; Min W Mid-Infrared Metabolic Imaging with Vibrational Probes. *Nat. Methods* 2020, 17 (8), 844–851. [PubMed: 32601425]
- (18). Centrone A Infrared Imaging and Spectroscopy Beyond the Diffraction Limit. *Annu. Rev. Anal. Chem* 2015, 8 (1), 101–126.
- (19). Yang H; Yang S; Kong J; Dong A; Yu S Obtaining Information about Protein Secondary Structures in Aqueous Solution Using Fourier Transform IR Spectroscopy. *Nat. Protoc* 2015, 10 (3), 382–396. [PubMed: 25654756]
- (20). SYu V; Prendergast FG Water (H₂O and D₂O) Molar Absorptivity in the 1000-4000 Cm⁻¹ Range and Quantitative Infrared Spectroscopy of Aqueous Solutions. *Anal. Biochem* 1997, 248 (2), 234–245. [PubMed: 9177749]

- (21). Lyng F; Gazi E; Gardner P In Biomedical Applications of Synchrotron Infrared Microspectroscopy, RSC Analytical Spectroscopy Monographs; Moss D, Eds; Royal Society of Chemistry Cambridge: Cambridge, U.K., 2011; No. 11, pp 147–185.
- (22). Hobro AJ; Smith NI An Evaluation of Fixation Methods: Spatial and Compositional Cellular Changes Observed by Raman Imaging. *Vib. Spectrosc* 2017, 91, 31–45.
- (23). Mohlenhoff B; Romeo M; Diem M; Wood BR Mie-Type Scattering and Non-Beer-Lambert Absorption Behavior of Human Cells in Infrared Microspectroscopy. *Biophys. J* 2005, 88 (5), 3635–3640. [PubMed: 15749767]
- (24). Bik E; Ishigaki M; Blat A; Jaształ A; Ozaki Y; Malek K; Baranska M Lipid Droplet Composition Varies Based on Medaka Fish Eggs Development as Revealed by NIR-, MIR, and Raman Imaging. *Molecules* 2020, 25 (4), 817. [PubMed: 32070018]
- (25). Spadea A; Denbigh J; Lawrence MJ; Kansiz M; Gardner P Analysis of Fixed and Live Single Cells Using Optical Photothermal Infrared with Concomitant Raman Spectroscopy. *Anal. Chem* 2021, 93 (8), 3938–3950. [PubMed: 33595297]
- (26). Zarei A; Klumbach S; Keppler H The Relative Raman Scattering Cross Sections of H₂O and D₂O, with Implications for In Situ Studies of Isotope Fractionation. *ACS Earth Space Chem.* 2018, 2 (9), 925–934.
- (27). Kuhar N; Sil S; Verma T; Umapathy S Challenges in Application of Raman Spectroscopy to Biology and Materials. *RSC Adv.* 2018, 8 (46), 25888–25908. [PubMed: 35541973]
- (28). Reffner J Advances in Infrared Microspectroscopy and Mapping Molecular Chemical Composition at Submicrometer Spatial Resolution. *Spectroscopy* 2018, 33 (9), 12–17.
- (29). Zhang D; Li C; Zhang C; Slipchenko MN; Eakins G; Cheng J-X Depth-Resolved Mid-Infrared Photothermal Imaging of Living Cells and Organisms with Submicrometer Spatial Resolution. *Sci. Adv* 2016, 2 (9), e1600521. [PubMed: 27704043]
- (30). Klementieva O; Sandt C; Martinsson I; Kansiz M; Gouras GK; Borondics F Super-Resolution Infrared Imaging of Polymorphic Amyloid Aggregates Directly in Neurons. *Adv. Sci* 2020, 7 (6), 1903004.
- (31). Schindelin J; Arganda-Carreras I; Frise E; Kaynig V; Longair M; Pietzsch T; Preibisch S; Rueden C; Saalfeld S; Schmid B; et al. Fiji: An Open-Source Platform for Biological-Image Analysis. *Nat. Methods* 2012, 9 (7), 676–682. [PubMed: 22743772]
- (32). Van Rossum G; Drake FL Python 3 Reference Manual; CreateSpace: Scotts Valley, CA, 2009.
- (33). Zebisch K; Voigt V; Wabitsch M; Brandsch M Protocol for Effective Differentiation of 3T3-L1 Cells to Adipocytes. *Anal. Biochem* 2012, 425 (1), 88–90. [PubMed: 22425542]
- (34). Derenne A; Claessens T; Conus C; Goormaghtigh E Infrared Spectroscopy of Membrane Lipids. In *Encyclopedia of Biophysics*; Roberts GCK, Ed.; Springer: Berlin, Heidelberg, 2013; pp 1074–1081.
- (35). Gazi E; Gardner P; Lockyer NP; Hart CA; Brown MD; Clarke NW Direct Evidence of Lipid Translocation between Adipocytes and Prostate Cancer Cells with Imaging FTIR Microspectroscopy. *J. Lipid Res* 2007, 48 (8), 1846–1856. [PubMed: 17496269]
- (36). Rak S; Zhan TD; Stefulj J; Kosović M; Gamulin O; Osmak M FTIR Spectroscopy Reveals Lipid Droplets in Drug Resistant Laryngeal Carcinoma Cells through Detection of Increased Ester Vibrational Bands Intensity. *Analyst* 2014, 139 (13), 3407–3415. [PubMed: 24834449]
- (37). Marecek J; Song B; Brewer S; Belyea J; Dyer RB; Raleigh DP A Simple and Economical Method for the Production of ¹³C,¹⁸O-Labeled Fmoc-Amino Acids with High Levels of Enrichment: Applications to Isotope-Edited IR Studies of Proteins. *Org. Lett* 2007, 9 (24), 4935–4937. [PubMed: 17958432]
- (38). Tadesse L; Nazarbachi R; Walters L Isotopically Enhanced Infrared Spectroscopy: A Novel Method for Examining Secondary Structure at Specific Sites in Conformationally Heterogeneous Peptides. *J. Am. Chem. Soc* 1991, 113 (18), 7036–7037.
- (39). Collins JM; Neville MJ; Pinnick KE; Hodson L; Ruyter B; van Dijk TH; Reijngoud D-J; Fielding MD; Frayn KN De Novo Lipogenesis in the Differentiating Human Adipocyte Can Provide All Fatty Acids Necessary for Maturation. *J. Lipid Res* 2011, 52 (9), 1683–1692. [PubMed: 21677304]

- (40). Muhamadali H; Chisanga M; Subaihi A; Goodacre R Combining Raman and FT-IR Spectroscopy with Quantitative Isotopic Labeling for Differentiation of E. Coli Cells at Community and Single Cell Levels. *Anal. Chem* 2015, 87 (8), 4578–4586. [PubMed: 25831066]
- (41). Shi J; Malik J Normalized Cuts and Image Segmentation. *IEEE Trans. Pattern Anal. Mach. Intell* 2000, 22 (8), 888–905.
- (42). Crase S; Hall B; Thennadil S Cluster Analysis for IR and NIR Spectroscopy: Current Practices to Future Perspectives. *Comput. Mater. Contin* 2021, 69 (2), 1945–1965.
- (43). Angel A; Desai K; Halperin ML Free Fatty Acid and ATP Levels in Adipocytes during Lipolysis. *Metabolism* 1971, 20 (1), 87–99. [PubMed: 5539064]
- (44). Layton C; Bancroft JD; Suvana SK 4 - Fixation of Tissues. In *Bancroft's Theory and Practice of Histological Techniques* (Eighth Edition); Suvana SK, Layton C, Bancroft JD, Eds.; Elsevier, 2019; pp 40–63.

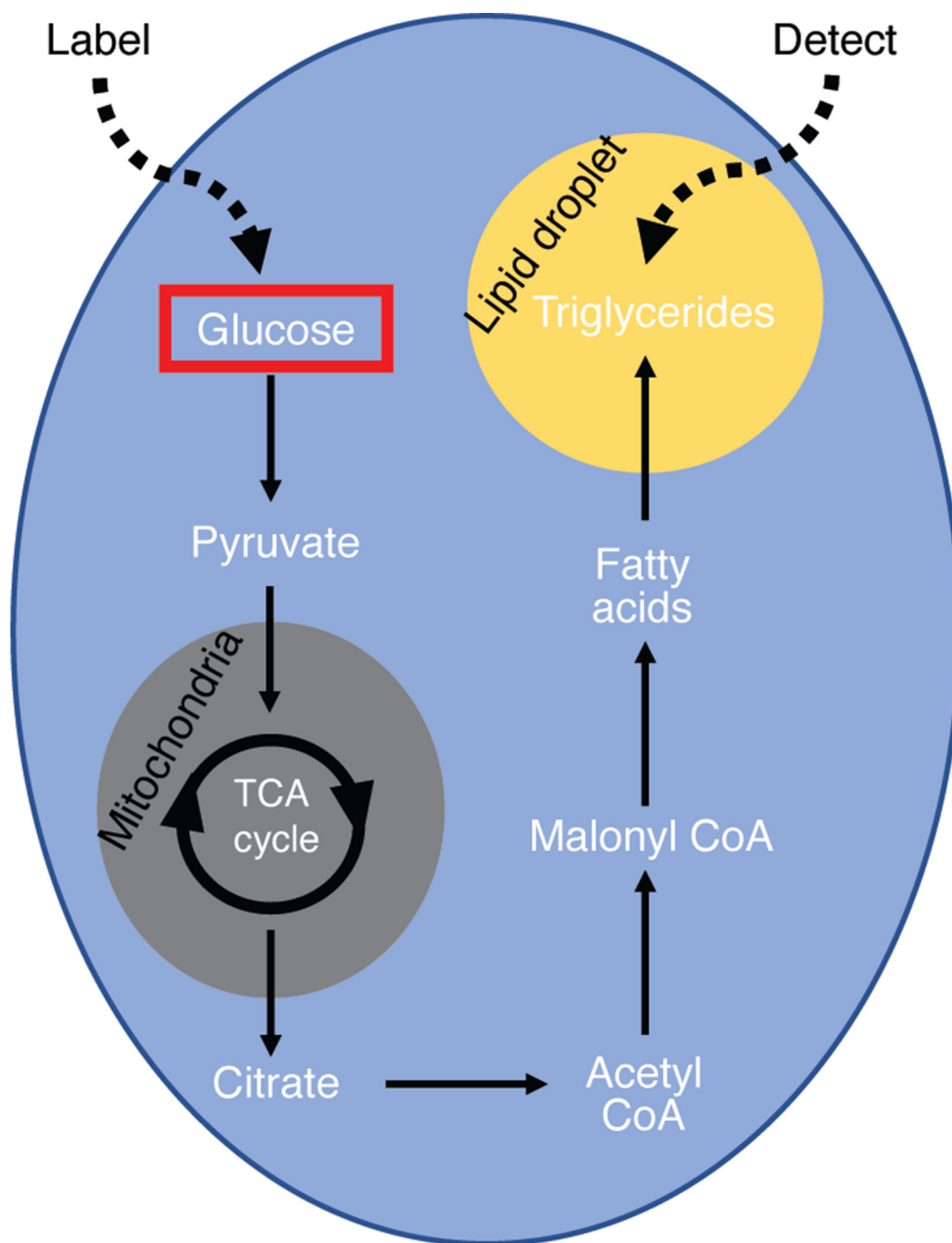
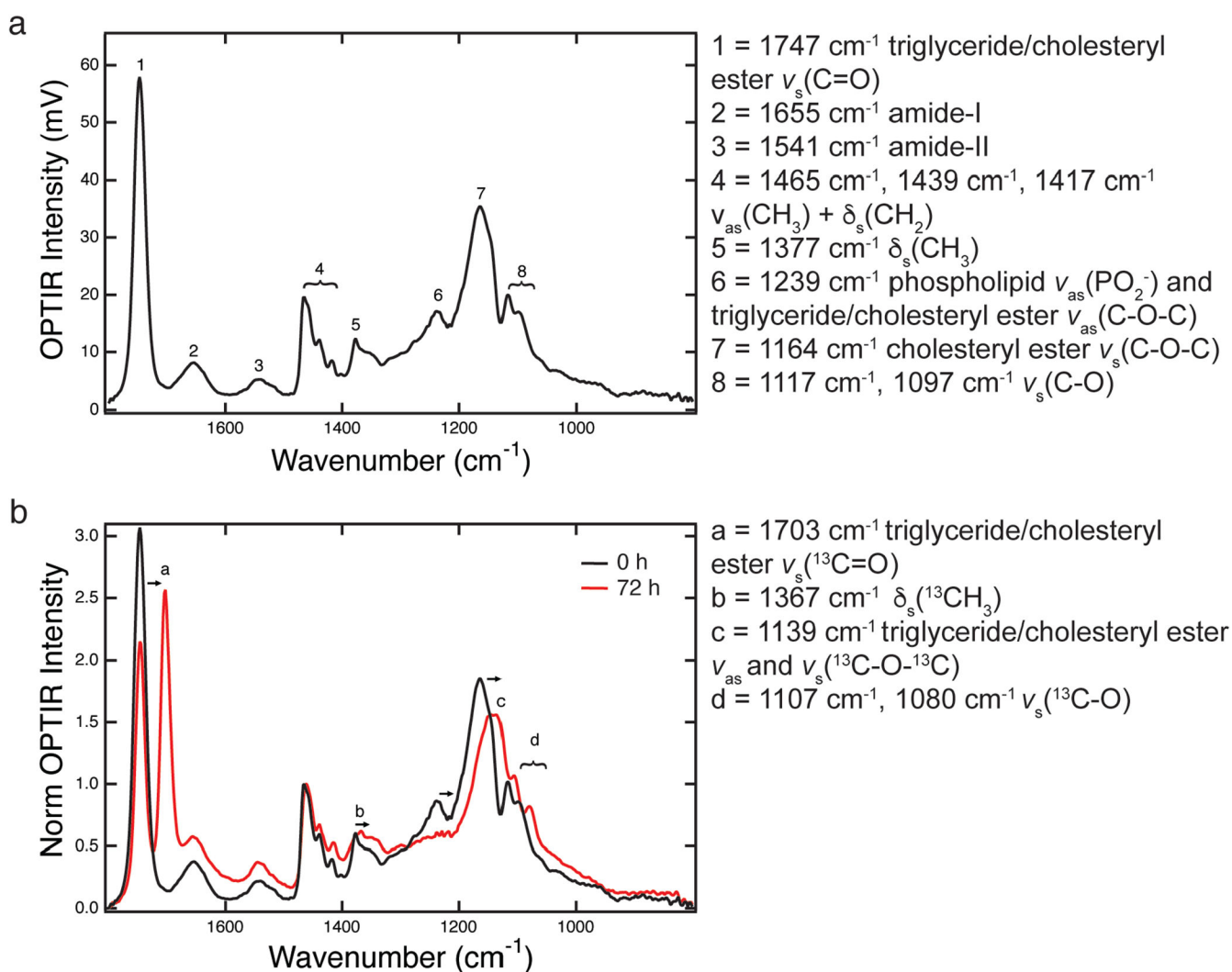


Figure 1.
De novo lipogenesis

**Figure 2.**

Peak assignments for ^{12}C and ^{13}C glucose fed 3T3-L1 adipocytes. (a) Representative OPTIR spectrum from a fixed, differentiated 3T3-L1 adipocyte with all peaks assigned. (b) Reprint of spectrum from (a) (black) overlaid with representative spectrum of differentiated 3T3-L1 adipocyte after 72 hour incubation with ^{13}C glucose. Peak shifts are indicated with arrows.

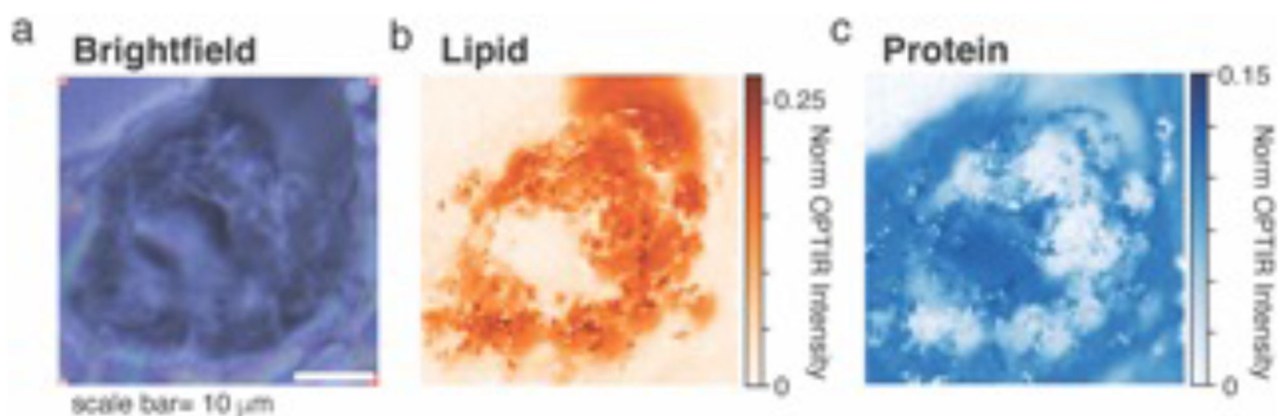


Figure 3. Images generated from OPTIR hyperspectral map of 3T3-L1 adipocyte. (a) Brightfield image of fixed and dried differentiated 3T3-L1 cell. Lipid ($1737 - 1757 \text{ cm}^{-1}$) (b) and protein ($1625 - 1700 \text{ cm}^{-1}$) (c) intensity maps of the same cell created by vector normalizing all spectra and taking average intensity over relevant spectral regions.

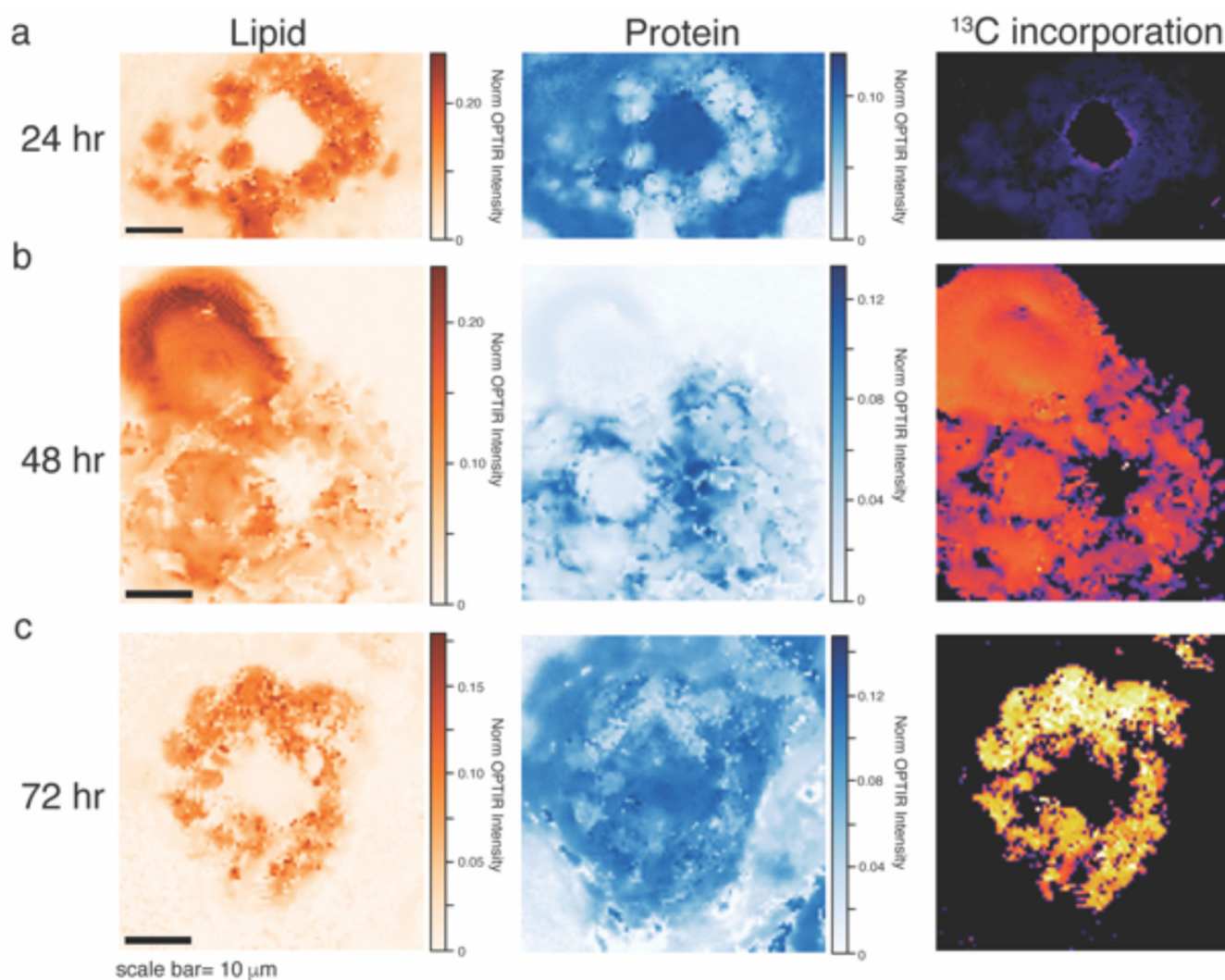


Figure 4.

Changes in lipid composition over time showcase rates of DNL. Images generated from the vector normalized and average intensity of ^{12}C lipid ($1737\text{-}1757\text{ cm}^{-1}$) (orange), protein ($1625\text{-}1700\text{ cm}^{-1}$)

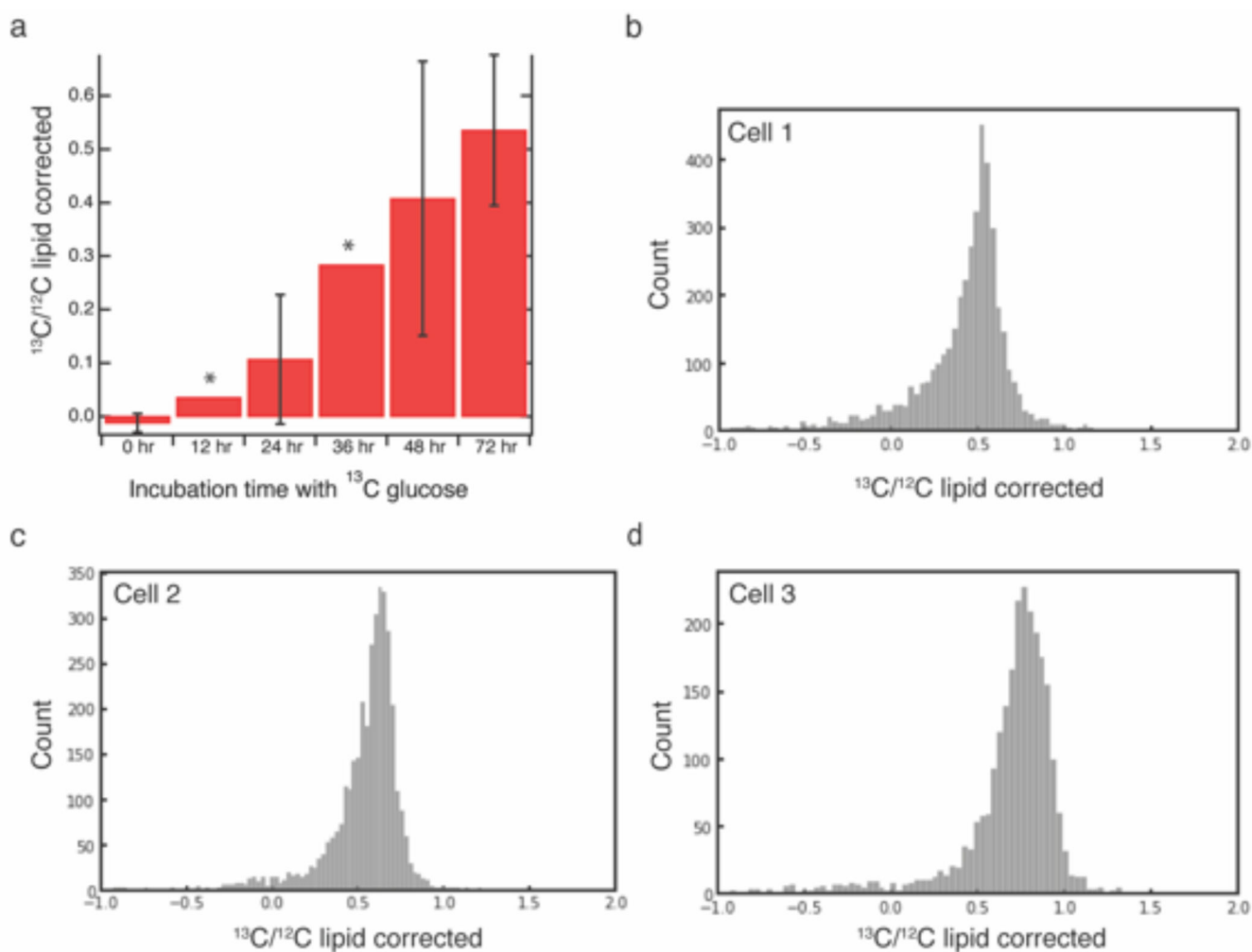


Figure 5.

Increasing ratio of ^{13}C labeled lipid over time. (a) Average ratio of $^{13}\text{C}:^{12}\text{C}$ lipid as calculated by eq. 1 of lipid rich regions of cell ($n = 3$ cells where error bars shown, * = only one cell measured for these time points). Between 1000 and 6000 spectra made up the lipid rich regions of each cell. (b-d) Histograms of $^{13}\text{C}:^{12}\text{C}$ lipid ratios for three cells at 72 hours after addition on ^{13}C glucose, demonstrating inter and intra-cellular spread.

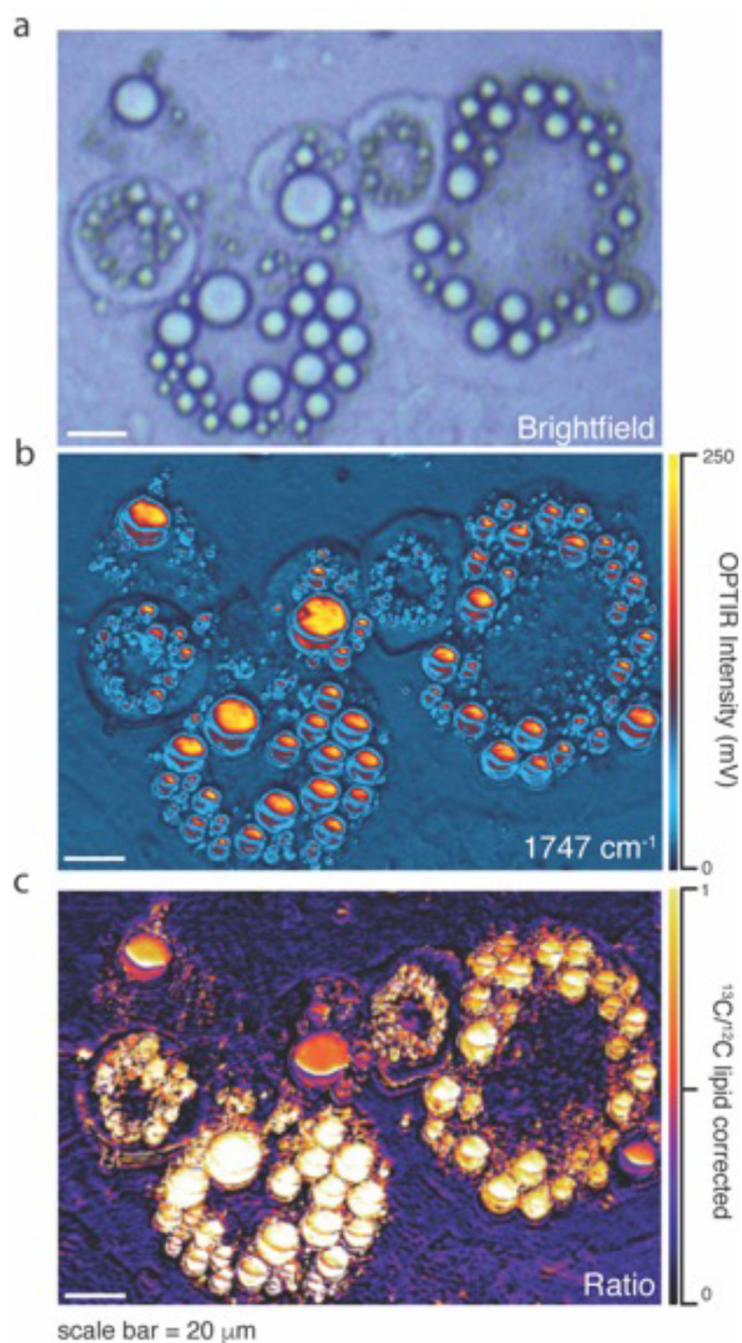


Figure 6.

Visualization of rates of DNL in live adipocytes. Live differentiated 3T3-L1 adipocytes 72 hours after feeding with ^{13}C glucose. (a) Brightfield image. (b) Single wavenumber image collected at 1747 cm^{-1} corresponding with ^{12}C lipid ester carbonyl band. (c) Ratio image showing ^{13}C lipid ester carbonyl (1703 cm^{-1}) / ^{12}C lipid ester carbonyl (1747 cm^{-1}) after correction for amide-I and water bending band highlighting varying rates of DNL across the cell. Data collected in PBS.

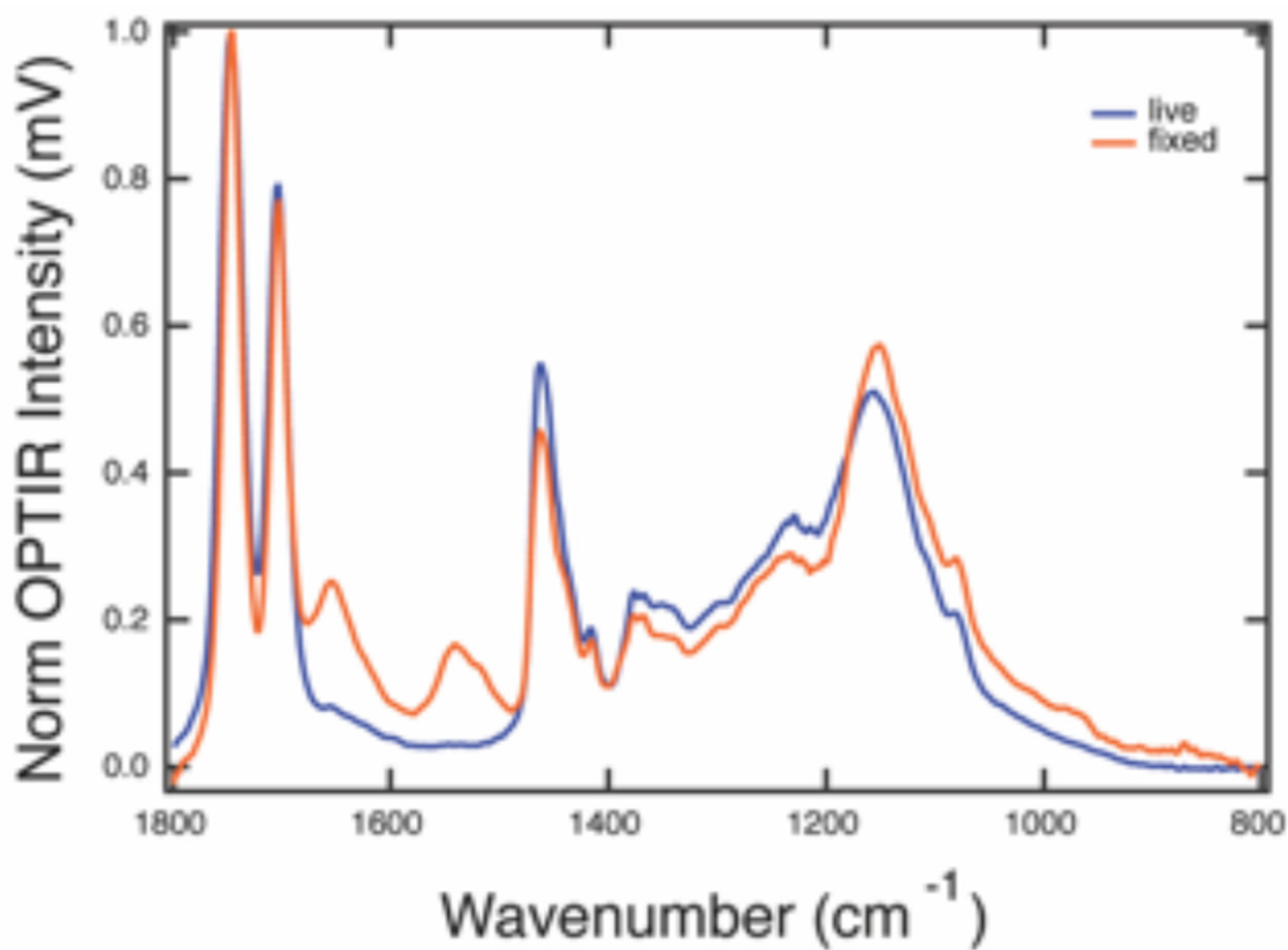


Figure 7.

Representative spectra from live and fixed cells. Spectra from live (blue) and fixed (orange) differentiated 3T3-L1 adipocytes 72 hours after feeding with ^{13}C glucose. Data normalized to ^{12}C lipid ester carbonyl.